Problem H3

Gamma Matrix Conventions & The Gordon Identity

Based on Peskin & Schroeder Problem 3.2

Step-by-step solution with prerequisite math review

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§1 The Problem Statement

H3 — Problem Statement (Reworded for Clarity)

Context: There are two standard explicit representations of the Dirac γ-matrices that satisfy the same Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$. Griffiths uses the **Dirac–Pauli representation**, where γ^0 is diagonal. Most QFT textbooks (Peskin & Schroeder, Lancaster & Blundell) use the **Chiral/Weyl representation** ("alt convention"), where γ^5 is diagonal instead. The two are related by swapping the roles of γ^0 and γ^5 (with a relative sign flip on γ^5); notably, the spatial matrices γ^i are identical in both conventions.

(a) Construct $\not{p}$ in each representation. Write out the matrix $\not{p} = \gamma^\mu p_\mu$ as an explicit 4×4 matrix (in terms of E and $\mathbf{p}$) using both the Griffiths and alt conventions. Compare the two results: which blocks are diagonal, which are off-diagonal, and why?

(b) Construct σ^{01} in each representation. The antisymmetric tensor of γ-matrices is defined by

$$\sigma^{\mu\nu} \equiv \frac{i}{2}[\gamma^\mu, \gamma^\nu] = \frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) \quad \text{(Griffiths Eq. 7.69)}$$

Compute σ^{01} explicitly as a 4×4 matrix in both conventions and compare.

(c) Prove the Gordon Identity. Show that for on-shell Dirac spinors (i.e., $u(p)$ and $u(p')$ satisfying the Dirac equation with mass m):

$$\bar{u}(p')\,\gamma^\mu\,u(p) = \bar{u}(p')\left(\frac{p^\mu + p'^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p'_\nu - p_\nu)}{2m}\right)u(p)$$

Method: Start from the right-hand side. Expand $\sigma^{\mu\nu}$ using its definition, simplify using the Clifford algebra anticommutation relation, and then use the Dirac equation ($\not{p}\,u(p) = m\,u(p)$ and $\bar{u}(p')\not{p}' = m\,\bar{u}(p')$) to collapse the result down to $\bar{u}(p')\gamma^\mu u(p)$.

Key constraint: The proof of part (c) should *not* use any explicit representation of the γ-matrices or spinors — only the Clifford algebra and the Dirac equation. This guarantees the identity holds in *any* convention.

§2 Math & Linear Algebra Review: What You Need

Before touching any physics, let's make sure the necessary mathematical tools are solid. This problem uses a specific set of linear algebra concepts — nothing beyond 4×4 matrix algebra, but you need to be fluent in it.

§2.1 Block Matrix Multiplication

Block Matrices

A 4×4 matrix can be written as a 2×2 grid of 2×2 blocks:

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

where A, B, C, D are each 2×2 matrices. Block multiplication works *exactly* like ordinary matrix multiplication, but with matrix entries:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{pmatrix}$$

Caution: The order of multiplication within each block matters! $AE \neq EA$ in general. This is essential because all γ-matrices are written in 2×2 block form using Pauli matrices.

§2.2 The Commutator and Anticommutator

Commutator & Anticommutator

For any two matrices A and B:

$$[A, B] \equiv AB - BA \qquad \text{(commutator)}$$

$$\{A, B\} \equiv AB + BA \qquad \text{(anticommutator)}$$

If $[A, B] = 0$, the matrices commute (order doesn't matter).
If $\{A, B\} = 0$, the matrices anticommute ($AB = -BA$).

A useful rearrangement: from the definitions, you can always write:

$$AB = \frac{1}{2}\{A, B\} + \frac{1}{2}[A, B]$$

This identity decomposes *any* product into its symmetric and antisymmetric parts. It's the key algebraic move in the Gordon Identity proof.

§2.3 The Trace (Reminder)

Trace Properties

$\mathrm{Tr}(M) = \sum_i M_{ii}$ (sum of diagonal elements). Key properties:

$$\mathrm{Tr}(A + B) = \mathrm{Tr}(A) + \mathrm{Tr}(B) \tag{1}$$

$$\mathrm{Tr}(AB) = \mathrm{Tr}(BA) \qquad \text{(cyclic property)} \tag{2}$$

$$\mathrm{Tr}(I_n) = n \qquad (n \times n \text{ identity}) \tag{3}$$

In particular, $\mathrm{Tr}(I_4) = 4$ (this is where the ubiquitous factor of 4 comes from).
A matrix is **traceless** if $\mathrm{Tr}(M) = 0$. All Pauli matrices and all Dirac γ-matrices are traceless.

§2.4 Pauli Matrix Algebra (Quick Reference)

Pauli Matrix Essentials

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The product rule that encodes everything:

$$\sigma_i \sigma_j = \delta_{ij}\, I_2 + i\varepsilon_{ijk}\, \sigma_k$$

Key consequence: $\boldsymbol{\sigma} \cdot \mathbf{a}$ is the 2×2 matrix:

$$\boldsymbol{\sigma} \cdot \mathbf{a} = a_x\sigma_1 + a_y\sigma_2 + a_z\sigma_3 = \begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix}$$

We will also need the "four-vector sigma" notation:

$$\sigma^\mu \equiv (I_2,\ \boldsymbol{\sigma}) = (I_2,\ \sigma_1,\ \sigma_2,\ \sigma_3)$$

$$\bar{\sigma}^\mu \equiv (I_2,\ -\boldsymbol{\sigma}) = (I_2,\ -\sigma_1,\ -\sigma_2,\ -\sigma_3)$$

These are *not* four-vectors in the Lorentz sense — they're shorthand for a set of four 2×2 matrices.

§2.5 The Dirac Equation as an Algebraic Constraint

The Dirac Equation — The Constraint That Powers the Proof

The free-particle Dirac equation in momentum space says two things:
Acting on a particle spinor from the left:

$$\not{p}\, u(p) = m\, u(p) \qquad (\text{i.e., } (\not{p} - m)\, u(p) = 0)$$

Acting on an adjoint spinor from the right:

$$\bar{u}(p)\, \not{p} = m\, \bar{u}(p) \qquad (\text{i.e., } \bar{u}(p)\, (\not{p} - m) = 0)$$

The crucial subtlety: **the momentum in $\not{p}$ must match the momentum of the spinor it acts on.**
For example, $\bar{u}(p')\, \not{p}' = m\, \bar{u}(p')$ is valid (both have p'). But $\bar{u}(p')\, \not{p} \neq m\, \bar{u}(p')$ because $\not{p}$ has momentum p, while the spinor has momentum p'. This mismatch is what makes the Gordon Identity proof nontrivial — some terms can be simplified immediately, and others require algebraic rearrangement first.

§3 The Two Conventions

§3.1 Griffiths Convention (Dirac–Pauli Representation)

Griffiths γ-Matrices

$$\gamma^0_{\mathrm{G}} = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \gamma^i_{\mathrm{G}} = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

$$\gamma^5_{\mathrm{G}} = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$$

Note: γ^0_{G} is diagonal, γ^5_{G} is off-diagonal.

§3.2 Alt Convention (Chiral/Weyl Representation)

Alt/Chiral γ-Matrices

The switch: what Griffiths calls γ^5 (up to a sign) becomes γ^0, and vice versa.

$$\gamma^0_{\mathrm{alt}} = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \qquad \gamma^i_{\mathrm{alt}} = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

$$\gamma^5_{\mathrm{alt}} = -\begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}$$

Note: γ^0_{alt} is off-diagonal, γ^5_{alt} is diagonal. The spatial matrices γ^i are the *same* in both conventions!

Why Two Conventions?

Both representations satisfy the *same* Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} I_4$. All physical results (cross sections, traces, etc.) are identical. The choice is a matter of convenience:
The Dirac–Pauli form is nice for non-relativistic limits (the upper two components of u dominate at low energy, because γ^0 is diagonal).
The Chiral/Weyl form is nice for massless particles and weak interactions (because γ^5 is diagonal, so left- and right-handed components live in the upper and lower halves of the spinor).

The Compact Notation

In the alt convention, all four γ^μ can be written in one clean formula:

$$\gamma^\mu_{\mathrm{alt}} = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}$$

where $\sigma^\mu = (I, \boldsymbol{\sigma})$ and $\bar{\sigma}^\mu = (I, -\boldsymbol{\sigma})$.
Check: for $\mu = 0$, $\sigma^0 = \bar{\sigma}^0 = I$, so $\gamma^0 = \left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$. ✓
For $\mu = i$, $\sigma^i = \sigma_i$ and $\bar{\sigma}^i = -\sigma_i$, so $\gamma^i = \left(\begin{smallmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{smallmatrix}\right)$. ✓
This compact notation is why textbooks prefer the alt convention — everything is written with σ^μ and $\bar{\sigma}^\mu$ instead of case-by-case.

§4 Part (a): $\not{p}$ in Both Conventions

Part (a): What We're Doing and Why

We want to see what $\not{p} = \gamma^\mu p_\mu = \gamma^0 E - \boldsymbol{\gamma} \cdot \mathbf{p}$ looks like as an explicit 4×4 matrix in each convention. The point is to *see* the structural difference: the same physical object ($\not{p}$) has a completely different matrix layout depending on which representation you use, even though its eigenvalues and algebraic properties are identical.
Recall that $p_\mu = (E, -\mathbf{p})$, so $\gamma^\mu p_\mu = \gamma^0 E - \gamma^1 p_x - \gamma^2 p_y - \gamma^3 p_z = \gamma^0 E - \boldsymbol{\gamma} \cdot \mathbf{p}$.

§4.1 Griffiths Convention

$$\begin{aligned} \not{p}_{\mathrm{G}} &= \gamma^0_{\mathrm{G}}\, E - \gamma^i_{\mathrm{G}}\, p_i \\ &= \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} E - \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} p_i \\ &= \begin{pmatrix} EI & 0 \\ 0 & -EI \end{pmatrix} - \begin{pmatrix} 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \\ -\boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{pmatrix} \end{aligned} \tag{4}$$

$\not{p}$ in Griffiths Convention

$$\boxed{\not{p}_{\mathrm{G}} = \begin{pmatrix} EI & -\boldsymbol{\sigma} \cdot \mathbf{p} \\ \boldsymbol{\sigma} \cdot \mathbf{p} & -EI \end{pmatrix}}$$

The diagonal blocks are $\pm E$ times the identity; the off-diagonal blocks involve $\boldsymbol{\sigma} \cdot \mathbf{p}$.

§4.2 Alt/Chiral Convention

$$\begin{aligned} \not{p}_{\mathrm{alt}} &= \gamma^0_{\mathrm{alt}}\, E - \gamma^i_{\mathrm{alt}}\, p_i \\ &= \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} E - \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} p_i \\ &= \begin{pmatrix} 0 & EI \\ EI & 0 \end{pmatrix} - \begin{pmatrix} 0 & \boldsymbol{\sigma} \cdot \mathbf{p} \\ -\boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{pmatrix} \end{aligned} \tag{5}$$

$\not{p}$ in Alt/Chiral Convention

$$\boxed{\not{p}_{\mathrm{alt}} = \begin{pmatrix} 0 & E\,I - \boldsymbol{\sigma} \cdot \mathbf{p} \\ E\,I + \boldsymbol{\sigma} \cdot \mathbf{p} & 0 \end{pmatrix} = \begin{pmatrix} 0 & p \cdot \bar{\sigma} \\ p \cdot \sigma & 0 \end{pmatrix}}$$

where $p \cdot \sigma = E\,I + \boldsymbol{\sigma} \cdot \mathbf{p}$ and $p \cdot \bar{\sigma} = E\,I - \boldsymbol{\sigma} \cdot \mathbf{p}$.
The diagonal blocks are **zero**; all the physics lives in the off-diagonal blocks. This is the hallmark of the chiral representation.

Comparison

In Griffiths, $\not{p}$ has nonzero diagonal blocks ($\pm E$). In the alt convention, $\not{p}$ has zero diagonal blocks and all the information is off-diagonal. Both encode the same physics — the eigenvalues of $\not{p}$ are the same ($\pm m$, each doubly degenerate), and $\not{p}^2 = p^2 I = m^2 I$ in both cases.

§5 Part (b): $\sigma^{\mu\nu}$ for $\mu = 0,\ \nu = 1$

Part (b): What We're Doing and Why

We need to compute one specific component of the tensor $\sigma^{\mu\nu}$ — namely σ^{01} — as an explicit 4×4 matrix in both conventions. This will reveal that even the "internal structure" of $\sigma^{\mu\nu}$ (which blocks are nonzero) depends on the representation, and it will give us practice with the commutator definition before we use $\sigma^{\mu\nu}$ abstractly in part (c).
The definition is:

$$\sigma^{\mu\nu} \equiv \frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$$

So $\sigma^{01} = \frac{i}{2}[\gamma^0, \gamma^1]$. We just need to multiply two 4×4 matrices and take their commutator.

What Is $\sigma^{\mu\nu}$?

The object $\sigma^{\mu\nu}$ is the **commutator** of two gamma matrices, scaled by $i/2$. Since a commutator is antisymmetric ($[A, B] = -[B, A]$), we have $\sigma^{\mu\nu} = -\sigma^{\nu\mu}$, so this tensor has 6 independent components (not 16). These correspond to the 6 generators of the Lorentz group (3 rotations + 3 boosts).
The combination $\bar{u}\,\sigma^{\mu\nu}\,u$ is the **tensor bilinear covariant** — it transforms as an antisymmetric rank-2 tensor under Lorentz transformations.

§5.1 Strategy

We need $\sigma^{01} = \frac{i}{2}[\gamma^0, \gamma^1] = \frac{i}{2}(\gamma^0\gamma^1 - \gamma^1\gamma^0)$.
Since γ^0 and γ^1 anticommute (from the Clifford algebra with $g^{01} = 0$):

$$\gamma^0\gamma^1 + \gamma^1\gamma^0 = 2g^{01}I = 0 \quad \Longrightarrow \quad \gamma^1\gamma^0 = -\gamma^0\gamma^1$$

Therefore:

$$\sigma^{01} = \frac{i}{2}(\gamma^0\gamma^1 - (-\gamma^0\gamma^1)) = \frac{i}{2}(2\gamma^0\gamma^1) = i\gamma^0\gamma^1$$

Simplification When $\mu \neq \nu$

Whenever $\mu \neq \nu$, the gamma matrices anticommute, so the commutator is just twice the product:

$$\sigma^{\mu\nu} = i\gamma^\mu\gamma^\nu \qquad (\mu \neq \nu)$$

This saves half the work — you only need one matrix product, not two.

§5.2 Griffiths Convention

Compute $\gamma^0_{\mathrm{G}}\,\gamma^1_{\mathrm{G}}$ using block multiplication:

$$\gamma^0_{\mathrm{G}}\,\gamma^1_{\mathrm{G}} = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}\begin{pmatrix} 0 & \sigma_1 \\ -\sigma_1 & 0 \end{pmatrix} = \begin{pmatrix} I\cdot 0 + 0\cdot(-\sigma_1) & I\cdot\sigma_1 + 0\cdot 0 \\ 0\cdot 0 + (-I)(-\sigma_1) & 0\cdot\sigma_1 + (-I)\cdot 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \sigma_1 \\ \sigma_1 & 0 \end{pmatrix}$$

σ^{01} in Griffiths Convention

$$\sigma^{01}_{\mathrm{G}} = i\gamma^0_{\mathrm{G}}\gamma^1_{\mathrm{G}} = i\begin{pmatrix} 0 & \sigma_1 \\ \sigma_1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}$$

This is $i\sigma_1$ placed in the off-diagonal blocks.

§5.3 Alt/Chiral Convention

Compute $\gamma^0_{\mathrm{alt}}\,\gamma^1_{\mathrm{alt}}$:

$$\gamma^0_{\mathrm{alt}}\,\gamma^1_{\mathrm{alt}} = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_1 \\ -\sigma_1 & 0 \end{pmatrix} = \begin{pmatrix} 0\cdot 0 + I\cdot(-\sigma_1) & 0\cdot\sigma_1 + I\cdot 0 \\ I\cdot 0 + 0\cdot(-\sigma_1) & I\cdot\sigma_1 + 0\cdot 0 \end{pmatrix}$$

$$= \begin{pmatrix} -\sigma_1 & 0 \\ 0 & \sigma_1 \end{pmatrix}$$

σ^{01} in Alt/Chiral Convention

$$\sigma^{01}_{\mathrm{alt}} = i\gamma^0_{\mathrm{alt}}\gamma^1_{\mathrm{alt}} = i\begin{pmatrix} -\sigma_1 & 0 \\ 0 & \sigma_1 \end{pmatrix} = \begin{pmatrix} 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \\ 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \end{pmatrix}$$

This is $\pm i\sigma_1$ placed in the **diagonal** blocks — the opposite structure from Griffiths.

Comparison of σ^{01}

In Griffiths: σ^{01} is off-diagonal in the 2×2 block structure.
In the alt convention: σ^{01} is block-diagonal, with $-i\sigma_1$ and $+i\sigma_1$ acting independently on the upper and lower two-component spinors (i.e., the left-handed and right-handed Weyl spinors).
This reflects the deeper fact: in the chiral basis, generators of Lorentz boosts act *separately* on left- and right-handed components, which is why this convention is preferred for the weak interaction.

§6 Part (c): The Gordon Identity — Setup

The Gordon Identity — What We Need to Prove

For two on-shell Dirac spinors $u(p)$ and $u(p')$ (both satisfying $(\not{p} - m)u = 0$ with the same mass m, but generally different momenta p and p'):

$$\bar{u}(p')\,\gamma^\mu\,u(p) = \bar{u}(p')\left(\frac{p^\mu + p'^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p'_\nu - p_\nu)}{2m}\right)u(p)$$

In words: the vector current $\bar{u}\gamma^\mu u$ can be decomposed into a piece proportional to the average momentum (the "convection current") and a piece involving $\sigma^{\mu\nu}$ contracted with the momentum transfer (the "spin current").

Part (c): Strategy Overview

We want to prove that the left-hand side ($\bar{u}(p')\gamma^\mu u(p)$) equals the right-hand side. The problem instructs us to **start from the RHS** and show it simplifies to the LHS.
The proof has three main moves:

1. **Expand** $\sigma^{\mu\nu}$ back into gamma matrices using its definition.
2. **Apply the Dirac equation** wherever $\not{p}$ or $\not{p}'$ acts directly on the appropriate spinor (i.e., $\not{p}\,u(p) = mu(p)$ and $\bar{u}(p')\not{p}' = m\bar{u}(p')$).
3. **Use the anticommutator** $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ to rearrange terms where the Dirac equation *can't* be applied directly (because $\not{p}$ is next to a γ^μ instead of next to a spinor).

The $(p + p')^\mu$ terms from Term 1 and Term 2 will cancel, leaving only $\bar{u}'\gamma^\mu u$.

§6.1 The Key Algebraic Identity

Before touching spinors, we need one purely algebraic fact. Consider the product $\gamma^\mu\gamma^\nu$. We can always decompose any matrix product into its symmetric and antisymmetric parts:

Decomposition of $\gamma^\mu\gamma^\nu$

$$\gamma^\mu\gamma^\nu = \frac{1}{2}\{\gamma^\mu, \gamma^\nu\} + \frac{1}{2}[\gamma^\mu, \gamma^\nu] = g^{\mu\nu}I + \frac{1}{2}[\gamma^\mu, \gamma^\nu]$$

But $[\gamma^\mu, \gamma^\nu] = -2i\sigma^{\mu\nu}$ by definition, so:

$$\boxed{\gamma^\mu\gamma^\nu = g^{\mu\nu}I - i\sigma^{\mu\nu}}$$

This is the master identity for the Gordon proof. It lets us trade a product of two gamma matrices for a metric tensor (a number) plus a $\sigma^{\mu\nu}$ (an antisymmetric tensor).

§7 Part (c): The Proof — Step by Step

§7.1 Step 1: Write Out the RHS

The RHS is:

$$\text{RHS} = \bar{u}(p')\left(\frac{p^\mu + p'^\mu}{2m}\right)u(p) + \bar{u}(p')\left(\frac{i\sigma^{\mu\nu}(p'_\nu - p_\nu)}{2m}\right)u(p)$$

Let's call these Term 1 and Term 2 and work on them separately.

§7.2 Step 2: Rewrite $\sigma^{\mu\nu}$ in Terms of Gamma Matrices

In Term 2, substitute $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu] = \frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)$:

$$\begin{aligned} i\sigma^{\mu\nu}(p'-p)_\nu &= i\cdot\frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)(p'_\nu - p_\nu) \\ &= -\frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)(p'_\nu - p_\nu) \\ &= -\frac{1}{2}\left[\gamma^\mu\gamma^\nu(p'_\nu - p_\nu) - \gamma^\nu\gamma^\mu(p'_\nu - p_\nu)\right] \end{aligned} \tag{6}$$

Now use the fact that $\gamma^\nu(p'_\nu - p_\nu)$ means $\gamma^\nu p'_\nu - \gamma^\nu p_\nu = \not{p}' - \not{p}$. Similarly for the other term:

$$i\sigma^{\mu\nu}(p'-p)_\nu = -\frac{1}{2}\left[\gamma^\mu(\not{p}' - \not{p}) - (\not{p}' - \not{p})\gamma^\mu\right] \tag{7}$$

§7.3 Step 3: Expand Everything

Distributing:

$$i\sigma^{\mu\nu}(p'-p)_\nu = -\frac{1}{2}\left[\gamma^\mu\not{p}' - \gamma^\mu\not{p} - \not{p}'\gamma^\mu + \not{p}\gamma^\mu\right] \tag{8}$$

So Term 2 becomes:

$$\begin{aligned} \text{Term 2} &= \frac{1}{2m}\cdot\left(-\frac{1}{2}\right)\bar{u}(p')\left[\gamma^\mu\not{p}' - \gamma^\mu\not{p} - \not{p}'\gamma^\mu + \not{p}\gamma^\mu\right]u(p) \\ &= -\frac{1}{4m}\bar{u}(p')\left[\gamma^\mu\not{p}' - \gamma^\mu\not{p} - \not{p}'\gamma^\mu + \not{p}\gamma^\mu\right]u(p) \end{aligned}$$

§7.4 Step 4: Apply the Dirac Equation

This is where the Dirac equation does the heavy lifting.

On the right side of the sandwich, $u(p)$ satisfies $\not{p}\,u(p) = m\,u(p)$.
On the left side, $\bar{u}(p')$ satisfies $\bar{u}(p')\,\not{p}' = m\,\bar{u}(p')$.

Applying the Dirac Equation — Which Terms Simplify?

The rule is simple: $\not{p}$ can be replaced by m **only when it acts directly on the matching spinor**. Let's check each of the four terms (remembering the spinors sitting on either side of the bracket):

1. $\bar{u}(p')\,\gamma^\mu\,\not{p}'\,u(p)$: The $\not{p}'$ is next to $u(p)$, not $u(p')$ — **can't simplify** (wrong spinor!).

2. $\bar{u}(p')\,\gamma^\mu\,\not{p}\,u(p)$: The $\not{p}$ acts directly on $u(p)$ from the left — **replace** $\not{p}\,u(p) \to m\,u(p)$. This gives $m\,\bar{u}(p')\gamma^\mu u(p)$.

3. $\bar{u}(p')\,\not{p}'\,\gamma^\mu\,u(p)$: The $\not{p}'$ acts on $\bar{u}(p')$ from the right — **replace** $\bar{u}(p')\not{p}' \to m\,\bar{u}(p')$. This gives $m\,\bar{u}(p')\,\gamma^\mu\,u(p)$.

4. $\bar{u}(p')\,\not{p}\,\gamma^\mu\,u(p)$: The $\not{p}$ is next to $\bar{u}(p')$, not $u(p)$ — **can't simplify** (wrong spinor!).

Terms 1 and 4 survive and require further work (Step 5). Terms 2 and 3 each produce $m\,\bar{u}'\gamma^\mu u$.

So terms 2 and 3 each give $m\,\bar{u}(p')\gamma^\mu u(p)$, while terms 1 and 4 remain as-is:

$$\text{Term 2} = -\frac{1}{4m}\bar{u}(p')\left[\gamma^\mu\not{p}' - m\gamma^\mu - m\gamma^\mu + \not{p}\gamma^\mu\right]u(p)$$

$$= -\frac{1}{4m}\bar{u}(p')\left[\gamma^\mu\not{p}' + \not{p}\gamma^\mu - 2m\gamma^\mu\right]u(p)$$

§7.5 Step 5: Use the Anticommutator on the Remaining Terms

Now we deal with $\gamma^\mu\not{p}'$ and $\not{p}\gamma^\mu$. Use the Clifford algebra:

$$\gamma^\mu\not{p}' = \gamma^\mu\gamma^\nu p'_\nu \quad \text{and} \quad \{\gamma^\mu,\gamma^\nu\} = 2g^{\mu\nu}$$

So $\gamma^\mu\gamma^\nu = 2g^{\mu\nu} - \gamma^\nu\gamma^\mu$, which gives:

$$\gamma^\mu\not{p}' = \gamma^\mu\gamma^\nu p'_\nu = (2g^{\mu\nu} - \gamma^\nu\gamma^\mu)p'_\nu = 2p'^\mu - \not{p}'\gamma^\mu$$

Similarly:

$$\not{p}\gamma^\mu = \gamma^\nu p_\nu\gamma^\mu = (2g^{\nu\mu} - \gamma^\mu\gamma^\nu)p_\nu = 2p^\mu - \gamma^\mu\not{p}$$

Now apply the Dirac equation to the remaining $\not{p}'$ and $\not{p}$ terms:

$$\bar{u}(p')\,[\gamma^\mu\not{p}']\,u(p) = \bar{u}(p')[2p'^\mu - \not{p}'\gamma^\mu]u(p) = 2p'^\mu\,\bar{u}(p')u(p) - m\,\bar{u}(p')\gamma^\mu u(p)$$

$$\bar{u}(p')\,[\not{p}\gamma^\mu]\,u(p) = \bar{u}(p')[2p^\mu - \gamma^\mu\not{p}]u(p) = 2p^\mu\,\bar{u}(p')u(p) - m\,\bar{u}(p')\gamma^\mu u(p) \tag{9}$$

§7.6 Step 6: Combine Everything

Substituting back into Term 2:

$$\text{Term 2} = -\frac{1}{4m}\left[2p'^\mu\,\bar{u}'u - m\,\bar{u}'\gamma^\mu u + 2p^\mu\,\bar{u}'u - m\,\bar{u}'\gamma^\mu u - 2m\,\bar{u}'\gamma^\mu u\right] \tag{10}$$

(where I'm abbreviating $\bar{u}' \equiv \bar{u}(p')$ and $u \equiv u(p)$ for space.)

Collect terms:

$$\text{Term 2} = -\frac{1}{4m}\left[2(p'^\mu + p^\mu)\bar{u}'u - 4m\,\bar{u}'\gamma^\mu u\right]$$

$$= -\frac{(p'^\mu + p^\mu)}{2m}\bar{u}'u + \bar{u}'\gamma^\mu u$$

§7.7 Step 7: Add Term 1 and Term 2

$$\begin{aligned}\text{RHS} &= \text{Term 1} + \text{Term 2} \\ &= \frac{(p'^\mu + p^\mu)}{2m}\bar{u}'u + \left(-\frac{(p'^\mu + p^\mu)}{2m}\bar{u}'u + \bar{u}'\gamma^\mu u\right) \\ &= \bar{u}(p')\,\gamma^\mu\,u(p) \qquad (11)\end{aligned}$$

The $(p'^\mu + p^\mu)$ terms cancel exactly, leaving only $\bar{u}(p')\gamma^\mu u(p)$.

Gordon Identity — Proved

$$\boxed{\bar{u}(p')\,\gamma^\mu\,u(p) = \bar{u}(p')\left(\frac{p^\mu + p'^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p'_\nu - p_\nu)}{2m}\right)u(p)} \quad \square$$

§8 Physical Meaning of the Gordon Identity

What Does It Mean?

The Gordon Identity decomposes the QED current $\bar{u}(p')\gamma^\mu u(p)$ into two physically distinct pieces:

The "convection current" term: $\dfrac{p^\mu + p'^\mu}{2m}\,\bar{u}'u$

This is proportional to the average four-momentum. It's what you'd get for a *spinless* charged particle — just charge carried along by the motion. The factor $\bar{u}'u$ is a Lorentz scalar.

The "spin current" (magnetic moment) term: $\dfrac{i\sigma^{\mu\nu}q_\nu}{2m}\,\bar{u}'u$

This involves the momentum transfer $q = p' - p$ contracted with the tensor $\sigma^{\mu\nu}$. It represents the contribution from the particle's *spin* — specifically, from its magnetic moment interacting with external fields. This term is what gives rise to the $g = 2$ prediction of the Dirac equation.

Why the Gordon Identity Matters

The Gordon Identity is the starting point for understanding the **anomalous magnetic moment** of the electron. When you compute loop corrections to the QED vertex, the correction modifies the $\sigma^{\mu\nu}q_\nu$ coefficient from $1/(2m)$ to $(1 + a)/(2m)$, where $a = \alpha/(2\pi) + \ldots$ is the anomalous magnetic moment. Schwinger's famous $g - 2$ calculation begins from precisely this decomposition.

§9 Representation Independence

The Punch Line of This Problem

Parts (a) and (b) show that the explicit matrices look very different between the two conventions. But the Gordon Identity (part (c)) was proved using **only**:

1. The Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$

2. The definition $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$
3. The Dirac equation $(\not{p} - m)u = 0$

None of these depend on which representation you use. The Clifford algebra is the *same* in both conventions. The Dirac equation holds in both conventions. So the Gordon Identity is true in both conventions — and in any other representation you might invent. This is the central lesson: physical identities follow from the algebra, not from the explicit matrices.

§10 Summary of Key Steps & Techniques

Technique	**Where It Was Used**
Block matrix multiplication	Computing $\not{p}$ and σ^{01} explicitly in parts (a) and (b)
$\gamma^\mu\gamma^\nu = g^{\mu\nu} - i\sigma^{\mu\nu}$ (anticommutator + commutator decomposition)	Setting up the Gordon proof — the master algebraic identity
Dirac equation: $\not{p}\,u = mu$, $\bar{u}\not{p}' = m\bar{u}$	Collapsing $\not{p}$ acting on spinors to the mass m
Clifford algebra: $\gamma^\mu \not{p}' = 2p'^\mu - \not{p}'\gamma^\mu$	Rewriting products where Dirac eq. can't be directly applied
Cancellation of $(p' + p)^\mu$ terms	The convection-current pieces from Term 1 and Term 2 cancel, leaving $\bar{u}\gamma^\mu u$

Problem H3, based on Peskin & Schroeder Problem 3.2.
Sources: Griffiths Ch. 7, Thomson Ch. 4&6, Lancaster & Blundell Ch. 36.